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RESEARCH ARTICLE

CFTformer: End-to-End Cross-Frame Multi-Object Tracking With Transformer

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ABSTRACT Multi-object tracking (MOT) has been at the center of numerous applications from autonomous vehicles (AVs) to surveillance and even retail analytics. Traditional MOT methods typically rely on motion-based and appearance-based similarity information to associate detections across frames. However, the new transformer attention-based approach to MOT has removed the need for complex post-processing steps, such as graph optimization, allowing for end-to-end query tracking across frames. While the new transformer-based approaches offer many advantages, in the majority of these models the temporal dimension of the sequence is only considered in either the iterative processing of the frames or the memory of the queries. The proposed cross-frame multi-object tracking transformer (CFTforrmer) aims to improve one of the challenging areas of tracking, the association across different frames in the temporal dimension. In the proposed approach, the temporal identities of the frames are included in the positional encoding of the patches. This approach allows the encoder-decoder to track the queries more efficiently across the frames. For this model, scalable deformable-attention layers were used to design the encoder and decoder to decrease the computational cost. CFTformer also employs the proposed attention-based trajectory refinement (ATR) scheme to improve the tracking performance in blurred frames. The three-dimensional positional encoding of the patches helps the proposed ATR module to better capture the trajectories of the queries and generate smoother predictions. Overall, the model was able to achieve 1.7% and 0.6% improvement in the identification F1 score (IDF1) metric on MOT17 and MOT20 datasets while having ∼0-15% lower number of identity switches, compared to other transformer-based approaches. More accurate tracking and lower identity switches make this algorithm more suitable to be used in the field of autonomous driving. To access model's performance in AV applications, the BDD100K dataset was utilized for training and evaluation where the proposed approach achieved a 1.9% improvement in the IDF1 compared to other transformer-based models.

INDEX TERMS CFTformer, multi-object tracking (MOT), transformer network, positional encoding, trajectory refinement.

I. INTRODUCTION

Object tracking is one of the fundamental tasks of computer vision that involves identifying and continuously tracking the location of one or multiple objects across a series of video frames. [1], [2]. It has numerous applications, including autonomous vehicles (AVs) for self-driving cars, pedestrian detection for collision prevention, surveillance systems for intrusion detection, and sports analytics to enhance athletes' performance, among many others [3]. Early techniques such as the Kalman filter, mean-shift algorithms, and optical flow laid the foundation of object tracking. However, these methods primarily relied on handcrafted features to characterize objects and often struggled with challenges such as significant appearance changes, occlusions, and real-time processing requirements [4].

The emergence of deep learning significantly altered the landscape of many computer vision tasks, including

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object tracking. Since then, convolutional neural networks (CNNs) have been widely used to learn robust features directly from pixel-level data, overcoming many limitations of traditional methods [5]. By employing deep learning models for effective object representation and position prediction, novel techniques such as Siamese networks and deep regression models set new benchmarks in object tracking performance [6], [7], [8].

Traditionally, object tracking has been categorized into two types, single-object tracking (SOT) and multi-object tracking (MOT). The focus of SOT task is on tracking the location and identity of a single object throughout a sequence of video frames. Methods such as the fully-convolutional Siamese network (SiamFC) and generic object tracking using regression networks (GOTURN), have demonstrated the progress in this domain, leveraging deep learning approaches to enhance the accuracy and performance of this task [6], [7]. In contrast, MOT task involves tracking multiple objects simultaneously, which in turn introduces certain new complexities such as association and identity preservation. Models such as simple online and real-time tracking with a deep association metric (DeepSORT) and fair multi-object tracking (FairMOT) have combined deep learning based appearance models and novel data association strategies to address MOT challenges, achieving major improvements in these scenarios [9], [10].

Despite recent advancements, object tracking still faces many challenges. Occlusions, scale variations, and real-time processing requirements are some of these challenges that remain significant obstacles in this domain. Current research on MOT methods primarily focuses on addressing these issues and improving the overall tracking performance by developing new data association techniques and exploring novel architectures [8].

In recent years, one architecture that has gained significant attention due to its high performance and versatility is the transformer network. Originally designed for natural language processing (NLP) tasks by Vaswani et al. [11] in 2017, it was later adapted to be used for computer vision tasks as well. This architecture captures long-range dependencies without relying on recurrent layers, typically found in traditional models such as recurrent neural networks (RNNs) and long short-term memory (LSTM) models.

The transformer model consists of an encoder and a decoder, both built using layers of multi-head self-attention and feed-forward neural networks [12]. The self-attention mechanism enables the model to focus on different parts of the input sequence to capture relationships between tokens regardless of their relative positions, while the multi-head self-attention mechanism allows the model to attend to information from multiple representation subspaces simultaneously. Unlike RNNs, transformers lack a sequential inductive bias, therefore positional encoding is added to the input embeddings to provide information about the position of the tokens in the sequence. Similarly, the decoder comprises identical layers but includes an additional multi-head cross attention layer that correlates the outputs to the encoded inputs [13].

Transformers have set new milestones in NLP applications, such as chatbots, text reasoning, and information retrieval. Bidirectional encoder representations from transformers (BERT) [14] and generative pre-trained transformer (GPT) [15] have demonstrated the effectiveness of this architecture when pre-trained on large text datasets. These models have achieved state-of-the-art (SOTA) results across various NLP tasks. Transformers have also been used for automatic speech recognition (ASR) and speech synthesis. Transformer-based ASR models [16] have shown improvements in converting spoken language into text by capturing relationships between distant tokens in a sequence. These architectures are particularly well-suited for tasks that require contextual understanding of the entire input.

The performance of transformer architectures scales well with larger datasets and model sizes, enabling them to make use of additional computational resources and data availability. However, since these models already demand high computational power, increasing the dataset size exacerbates this issue. Recent advancements in transformer networks have focused on improving their efficiency and extending their applicability. Models such as sparse transformers [17] and efficient transformers [18] aim to reduce the computational complexity of the self-attention mechanism, enabling training on even larger datasets.

In recent years, transformers have been applied in more complex computer vision tasks such as object tracking, achieving SOTA results by utilizing the attention mechanism to capture contextual relationships and associate objects across frames [1]. They have improved the performance and robustness of tracking by providing a unified framework that integrates detection and tracking phases, eliminating the need for separate post-processing steps and complex heuristics employed by traditional methods [19], [20].

In this work, a new end-to-end attention-based MOT model, built on the transformer architecture of multi-scale deformable detection transformer (DETR) is presented. The proposed, cross-frame tracking transformer (CFTformer) tracks queries using their encoded temporal positions in addition to their spatial positions across the frames. By incorporating the temporal dimension of frames to the positional encoding, the encoder learns the relative frame positions of queries more effectively, enabling more accurate tracking across frames. Additionally, a new attention-based trajectory refinement (ATR) module is introduced that makes use of the new positional embeddings to refine query trajectories and enhance the prediction accuracy of the proposed tracker.

II. RELATED WORKS

Following the introduction of transformers to the field of computer vision with the release of models such as vision transformer (ViT) in 2020 [21], researchers have used this architecture to develop new models for object detection and

object tracking. One of the most notable approaches in object detection was DETR, which has been widely utilized in developing tracking models. Introduced by Carion et al. [22], DETR reformulated object detection as a direct set prediction problem, eliminating the need for hand-designed components such as anchor generation and non-maximum suppression. However, its core innovation lies in its use of the transformer encoder-decoder structure, allowing this model to capture global context and correlations within the images. The encoder processed the high-level image features extracted by a CNN backbone; while the decoder associated object bounding boxes and class labels through the cross-attention mechanism. This approach not only simplified the detection pipeline but also enhanced the model's ability to handle complex scenarios with overlapping objects, achieving SOTA performance on benchmark datasets.

In object tracking, recent transformer-based models have demonstrated the effectiveness of transformer architecture by integrating global context from multiple frames to manage complex tracking scenarios [20], [23], [24]. These models utilized the attention mechanism to maintain a comprehensive understanding of the scene improving the overall performance of both SOT and MOT tasks. In 2020, Sun et al. [23] introduced TransTrack, the first transformer-based MOT model. They designed this model using a transformer encoder-decoder framework where object queries interact with frame features extracted by a CNN backbone. The integration of transformer architecture enabled this model to achieve significant improvements in tracking performance on standard datasets [23].

Following TransTrack, most transformer-based MOT models have focused on integrating detection and association within a single training framework. Their goal is to improve the tracking performance by developing end-to-end architectures that minimize the need for multi-modular approaches and post-processing steps. In these cases, the transformer decoder directly manages the bounding box generation and intra-frame association. These approaches are typically based on the deformable DETR model. Deformable DETR [23] combines the concept of deformable convolutional networks with transformer architecture to accommodate varying object scales and complex motion patterns while improving the computational costs of the transformer algorithm. While these trackers excel at associating new objects with existing ones in the short term, some scenarios require extended temporal modeling. TrackFormer, developed by Meinhardt et al. [20], was one of these models. It correlated the spatial information through the self-attention mechanism while incorporating temporal information via the cross-attention mechanism to track the queries across the frames. Despite having a simple architecture, this model utilized the advantages of transformer models to improve tracking accuracy [20]. Similar to TrackFormer, the multi-object tracking with transformer (MOTR), introduced by Zheng et al. [24], utilized transformer architecture for both the detection and tracking tasks, combining them into a single end-to-end model. This approach simplified the tracking pipeline and improved the overall performance [24]. Zhang et al. [25] continued their work on end-to-end models and later published MOTRv2, an improved version of the original MOTR model. In this new model, they sought to further improve the performance by integrating YOLOX, a type of you only look once (YOLO) model, as the detector to generate query anchors. Likewise, contrastive transformer (ContrasTR) published by De Plaen et al. [26] took advantage of the features generated by DETR to generate discriminative identity-level representations that facilitate object re-identification. ContrasTR achieved this with minimal additional computation by employing instance-level contrastive loss and a more efficient sampling algorithm. The existing DETR model was extended with the contrastive module, which synchronized with the bounding box predictions and object queries to further optimize the association phase of the tracking. Another example of this group is the anchor-based query-tracking model (ABQ-Track). This model used an anchor-based query system, where queries are initialized at static spatial points and dynamically updated based on the movement of the object across the frames. It also included a template-matching module, which stores a reference template for each object. These templates maintain the appearance history of objects and are updated throughout the tracking process. The transformer framework processed the input frames and templates through layers of self-attention and cross-attention. Self-attention captured global spatial dependencies in the current frame, while cross-attention associated the current frame's features with object templates to ensure identity consistency.

The second type of transformer-based models integrates transformers with memory modules or graph structures to track objects over extended periods. These models specifically address challenges in MOT, such as occlusion and identity switching. However, this approach typically incurs higher computational costs and requires more extensive training. TransMOT proposed in 2023 by Chu et al. [27] is an example of this type of models. This framework utilized graph transformers to model the spatial and temporal interactions of objects with high accuracy. The combination of spatial-temporal modeling and graph-based attention mechanisms enabled TransMOT to achieve high performance in MOT tasks. It employed a graph transformer to compute and simulate relationships between objects in space and time, where nodes represented objects and edges depicted their interactions across frames. This model leveraged the self-attention mechanism to capture intra-frame and inter-frame dependencies, effectively measuring object relationships over time. As a result, TransMOT significantly reduced identity switching and improved tracking predictability. Another example of this group is the diffusion model for multi-object tracking (DiffusionTrack). This model employed a generative pipeline based on the concept

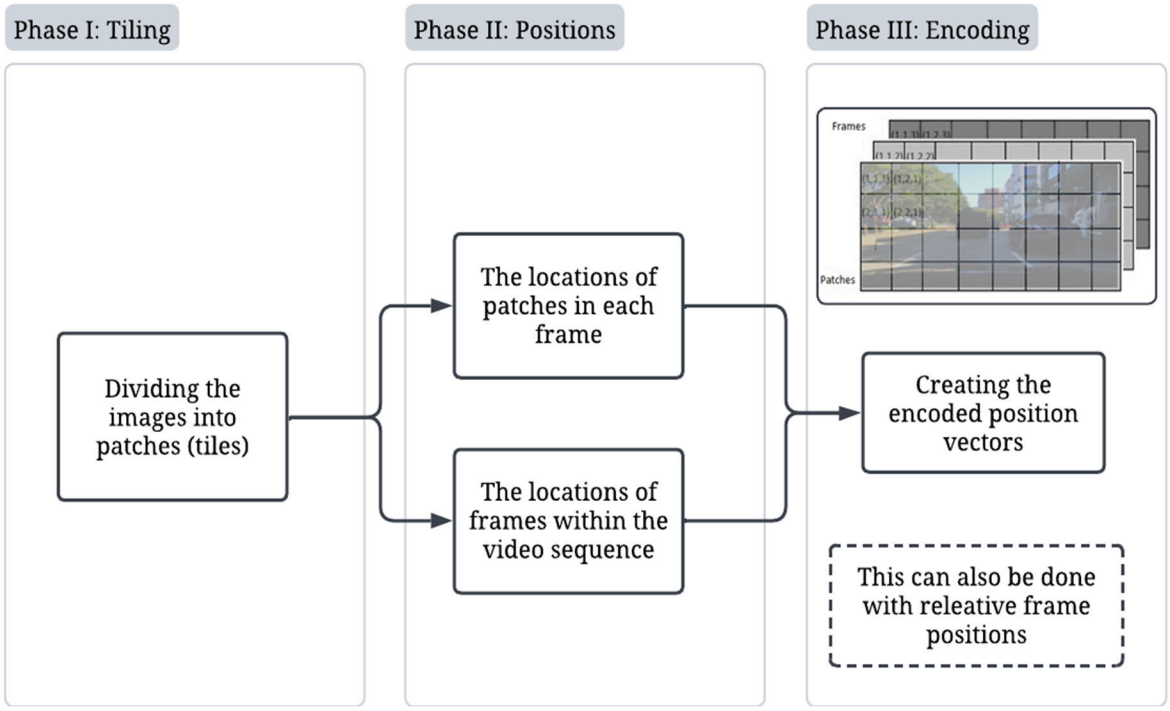


FIGURE 1. The three-dimensional positional encoding including the frame location in the video sequence.

of DiffusionDet [28], which incrementally refines a noisy initial prediction into an accurate, stable state through diffusion runs. It used a YOLOX model as the backbone to extract high-level features, which were then passed to the diffusion head. The diffusion head module consisted of an RoI pooler, a self-attention mechanism, and a newly designed spatial-temporal fusion (STF) sub-module. This approach enabled the model to separate useful signals from uncertain or overlapping detections, ensuring identity consistency and minimizing identity switching. Similarly ETTrack developed by Han et al. [29] introduced a novel approach by leveraging historical motion patterns of queries to predict accurate object movements. It merged the transformer architecture with a temporal convolutional network (TCN) to evaluate both long-term and short-term object movement patterns. The integrated model predicted object movements using trajectories from previous frames.

The proposed CFTformer incorporates both approaches. This model is an end-to-end framework based on the deformable DETR architecture; however, it also embeds the temporal information of patches within the queries. Additionally, the proposed trajectory refinement phase functions as a memory module, updating queries by leveraging historical motion information.

The use of transformers in MOT applications is a relatively new field and presents several unresolved challenges including high computational time and memory usage, which scale with sequence length adding up to substantial amounts. This can be problematic for real-time applications, where fast computing is critical, especially given the continuous increase in the resolution and frame rates of modern video recording devices [23]. Additionally, maintaining object identities across frames remains a challenge, especially in crowded scenes with rapid movement and high likelihood of occlusion [27]. These issues can further complicate the development of new transformer-based trackers [20]. Despite all these challenges, the ability of transformer models to capture global context regardless of the patch locations, as well as their resistance to gradient vanishing offers a significant potential that cannot be overstated.

III. PROPOSED APPROACH

Object tracking can be formulated as a mathematical problem where the objective is to estimate the trajectory of the objects over time in a sequence of frames. This task can be approached by defining a transfer function that maps the information of each video frame t to the positions and identities of the objects in that frame. Therefore, given a sequence of frames $\{I_t\}_{t=1}^{T}$ the goal is to generate a set of bounding boxes $\{B_t^i\}$ with their corresponding identities $\{ID_t^i\}$ for each object i in every frame t [30]. The primary challenge is to minimize the switching of identities caused by occlusions or changes in the object's appearance. This objective can be expressed as an optimization problem that seeks to minimize an overall cost function which penalizes identity switches and missed detections [31]. This formulation provides a rigorous framework for developing and evaluating tracking algorithms.

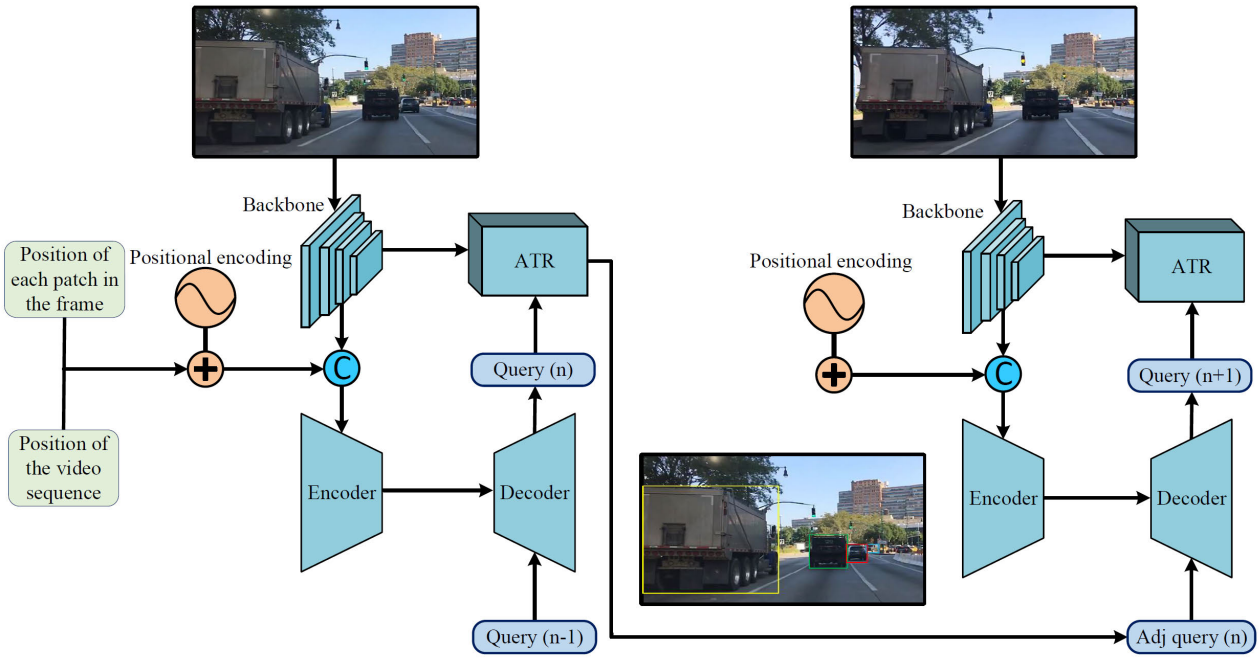


FIGURE 2. The architecture of the proposed CFTformer model including the details on the proposed positional encoding and the internal structure of the encoder/decoder based on the multi-scale multi-head deformable-attention mechanism.

CFTformer introduces a new approach to attention-based tracking by combining the temporal positions of frames with the spatial locations of patches to create hybrid positions. These hybrid positions are embedded within patches, preserving both their temporal order and spatial information. This helps the model maintain the contextual information of frames. Fig. 1 presents an overview of the proposed positional encoding approach. After the initial objects are detected, they interact with encoded features from previous frames during the tracking phase through self-attention mechanisms, enabling the attention module to learn temporal correlations between the current frame and patches from previous frames, thereby improving prediction accuracy. The queries then pass through the ATR module, where their correlations and dependencies with all previous queries are computed using a separate attention module. This correlation is subsequently used to refine the queries, ensuring improved tracking performance. While this approach effectively handles complex scenes, it requires greater computational resources and extensive training data. The following sections provide a detailed discussion of the internal structure of the CFTformer model.

A. FEATURE EXTRACTION

To extract the high-level features from the preprocessed frames, a backbone model is required. For CFTformer, a residual network (ResNet) with 50 layers is used, specifically ResNet-50 [33] as the backbone network. ResNet models are well known for their effectiveness in feature extraction, particularly in computer vision tasks. Among models with varying numbers of residual blocks, ResNet-50 is recognized for its high efficiency and has been widely used in previous tracking-by-attention models. This network mitigates the degradation problem in deep networks by integrating residual layers. Its architecture enables the network to learn identity mappings, which are added to the outputs of stacked layers, effectively facilitating the training of much deeper networks. Due to its ability to maintain high performance with relatively low computational cost compared to deeper networks, ResNet-50 is an ideal choice for the backbone algorithm. It processes input images through multiple convolutional layers, each designed to capture increasingly abstract features, ranging from low-level edges and textures to high-level semantic concepts such as object parts and scenes [33].

In CFTformer, the ResNet-50 backbone processes the pixel data of each frame to generate the feature maps that capture various characteristics of the objects and their surroundings. These feature maps contain detailed information, and serve as the fundamental data for subsequent processing steps.

B. QUERY INITIALIZATION

Parallel to the backbone, an object detection module identifies and classifies potential objects in each frame. During this phase, initial bounding boxes and class labels are generated, and unique IDs are assigned to each object. This step is

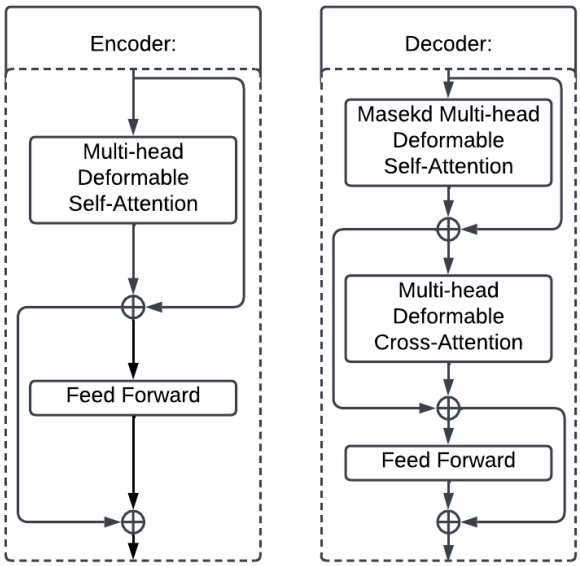


FIGURE 3. The encoder and decoder of CFTformer are based on multi-head deformable-attention mechanism.

crucial in guiding subsequent stages of the tracking process. Similar to other transformer-based approaches, CFTformer integrates the initialization and tracking phases by defining two distinct types of queries, one for static objects and the other for dynamic tracks. Static queries are used to detect new objects in each frame, and to provide the initial information about their locations and identities. These queries are independently initialized before tracking begins and remain constant throughout the procedure. Conversely, dynamic queries are continuously updated and track existing objects across frames. These track queries are initialized based on the outputs of the static queries and are updated as the objects move through the frames. The use of both static and dynamic queries allows the model to efficiently manage the initialization of new tracks while maintaining the continuity of existing ones. By leveraging the attention module of the transformer, these queries focus on relevant features in each frame, enabling the model to accurately detect and track the objects across sequences.

C. POSITIONAL ENCODER

Most transformer models used in computer vision, for both detection and tracking scenarios, rely on either a two-dimensional sinusoidal positional encoder or a learnable positional encoder. In detection, a two-dimensional positional encoding is sufficient to describe the location of queries. However, in tracking, with the addition of the time axis, traditional positional encodings cannot describe query locations across frames. As a result, the transformer model struggles to differentiate between patch locations, receiving only partial positional information through the recurrent query input of the decoder while remaining otherwise unaware of frame positions. Current transformer-based tracking methods are mostly designed based on object detection algorithms such as deformable DETR and mostly use the same positional encoding approach.

$$PE_{(x,y,f,2i)} = \sin\left(\frac{x}{10000^{2i/D}}\right) + \sin\left(\frac{y}{10000^{2i/D}}\right) + \sin\left(\frac{f}{10000^{2i/D}}\right) \quad (1)$$

$$PE_{(x,y,f,2i+1)} = \cos\left(\frac{x}{10000^{2i/D}}\right) + \cos\left(\frac{y}{10000^{2i/D}}\right) + \cos\left(\frac{f}{10000^{2i/D}}\right) \quad (2)$$

In two-dimensional approaches, patch positions are measured along two axes. By extending the patch information to the frame dimension, the three-dimensional sinusoidal positional encoding can be described as (1) and (2).

While the first two dimensions, corresponding to the x and y axes of the frames, are static, the third dimension, representing the frame sequence, is temporal. Unlike the spatial dimensions, frames in the temporal axis have a relative positional relationship. Therefore, the positional encoding term of the temporal axis can be substituted by (3) and (4) to create a hybrid positional encoding.

$$RPE_{(\Delta pos,2i)} = \sin\left(\frac{\Delta pos}{10000^{2i/D}}\right) \quad (3)$$

$$RPE_{(\Delta pos,2i+1)} = \cos\left(\frac{\Delta pos}{10000^{2i/D}}\right) \quad (4)$$

To implement this algorithm, at each step, previously processed frames are added to the current frame with a new relative positional encoding. These overlapping frame segments help reduce identity switches and provide additional context to the transformer model for predicting object queries. This novel approach of embedding frame positions within patches is the basis for the CFTformer model's name. By incorporating relational information about frames, it enhances tracking efficiency. Additionally, it contributes valuable data for trajectory refinement, as discussed in the following sections. While this method increases computational load and training time, it improves tracking performance and reduces the number of identity switches in return.

D. ENCODER/DECODER

In this step, the embedded feature vector is generated by combining the positional vector of each token with the features extracted by the backbone. The embedded features are then passed to the transformer encoder where the self-attention module processes them to capture relationships between different patches of the current and previous frames [11]. This step is crucial for understanding the spatial and contextual dynamics of the scene. The encoder of the CFTformer model utilizes the multi-scale deformable attention module, which focuses on relevant regions of an image at multiple scales and resolutions. This approach is specifically designed to handle

variations in object size and shape making it highly effective for tracking.

$$MSDeformAttn\left(z_q, \widehat{p_q}, \left\{x^l\right\}_{l=1}^{L}\right) = \sum_{m=1}^{M} W_m \left[\sum_{l=1}^{L}\sum_{k=1}^{K} A_{mqk}.W'_m x^l\left(\phi_l(\widehat{p_q}) + \Delta p_{mlqk}\right)\right] \tag{5}$$

In (5), $\left\{x^l\right\}_{l=1}^{L}$ represents the multi-scale feature map for x^l, $\widehat{p_q}$ denotes the location of the reference point and $\phi_l(\widehat{p_q})$ corresponds the position of the rescaled feature map for the *l-th* level. After the 'add and norm' step, the output of the attention layer passes through the feed-forward network before being forwarded to the decoder module. Fig. 2 provides a visual illustration of the overall model.

The decoder in CFTformer is a critical component that transforms encoded features into meaningful predictions for object tracking. Utilizing the transformer architecture, it operates by attending to both the encoded features and a set of current and past queries, which represent potential object tracks. The designed decoder employs masked multi-scale deformable self-attention to dynamically focus on relevant regions of the input, capturing spatial and temporal relationships between objects across frames. It then applies multi-scale deformable cross-attention to establish tracks for moving objects. Each track query is updated iteratively, allowing the decoder to refine its predictions for object locations, identities, and trajectories. The decoder utilizes the proposed positional encoding to associate queries with their relevant frame position within the video sequence. This process is performed iteratively, where the output queries from each frame serve as input queries for future frames, creating a unified integration with time domain information. By leveraging the self-attention and cross-attention modules, the decoder maintains object identities more consistently, even in complex and crowded scenes. This capability is particularly effective in handling occlusions and interactions between objects. Fig. 3 illustrates the internal structure of the encoder and decoder, including the deformable self-attention and cross-attention mechanisms used in the proposed CFTformer model.

E. MATCHING AND ASSOCIATION

After the decoding step, the proposed CFTformer algorithm performs a matching and association step to link detected objects across a sequence of frames. This process involves comparing predicted bounding boxes, their locations, and their identities with detections from previous frames to establish correspondences. To accomplish this, bipartite matching and the Hungarian algorithm are employed.

1) BIPARTITE MATCHING

In this step, the tracking problem is modeled as a bipartite graph where one set of vertices represents detected objects in the current frame, while the other set represents tracked objects from the previous frame. Edges between vertices from both sets are assigned a cost, typically reflecting variations between corresponding objects. The objective is to find a matching that minimizes the total cost across the graph, ensuring the correct association of each detected object in the current frame with its corresponding tracked object from the previous frame. The goal of bipartite matching can be calculated as the following metric.

$$\widehat{\sigma} = argmin_{\sigma\in\psi_N} \sum_{i}^{N} \mathcal{L}_{match}\left(y_i, \widehat{y}_{\sigma(i)}\right) \tag{6}$$

$$\mathcal{L}_{match} = -1_{\{c_i\neq\emptyset\}}\widehat{p}_{\sigma(i)}(c_i) + 1_{\{c_i\neq\emptyset\}}\mathcal{L}_{box}\left(b_i, \widehat{b}_{\sigma(i)}\right) \tag{7}$$

In (6) and (7), ψ_N represents the set of all permutation functions on natural number from 1 to N, $\widehat{y}_{\sigma(i)}$ denotes the *i-th* prediction, y_i is the ground truth c_i is the target class label and L_{box} represents the bounding box loss [22].

2) HUNGARIAN MATCHING

The Hungarian algorithm is a widely used approach for solving the bipartite matching problem in object tracking, valued for its efficiency and accuracy. It provides an optimal solution to the assignment problem while minimizing the overall matching cost. The Hungarian algorithm helps MOT systems handle occlusions and dynamic changes. Once the bipartite matching is computed, the Hungarian loss can be determined using (8) and (9) [22].

$$\mathcal{L}_{Hungarian} = \sum_{i=1}^{N}\left[-\log\widehat{p}_{\widehat{\sigma}(i)}(c_i) + 1_{\{c_i\neq\emptyset\}}\mathcal{L}_{box}\left(b_i, \widehat{b}_{\sigma(i)}\right)\right] \tag{8}$$

$$\mathcal{L}_{total}(y, \widehat{y}) = \sum_{i=1}^{N}\mathcal{L}_{query}(y_i, \widehat{y}_{\sigma(i)}) \tag{9}$$

Equations (8) and (9) illustrate how the Hungarian loss is calculated, where ψ_N, $\widehat{y}_{\sigma(i)}$ and L_{box} are similar to (6) and (7). Minimizing the total association cost ensures that the most likely correspondences are selected, even in crowded scenes. The total prediction loss of the model, shown in (9) is computed as the sum of losses for both tracking and detection of the queries.

F. TRAJECTORY REFINEMENT

In real-world object tracking, rapid object motion can cause frame blurring, which is a common issue that significantly impacts tracking performance [33]. This effect is especially pronounced in low-quality videos and those with lower frame rates. In such cases, trajectory refinement is needed to improve tracking accuracy. Beyond low-resolution samples, trajectory refinement is particularly effective in handling occlusions, ensuring that object trajectories remain intact even when objects are temporarily blocked. This process involves smoothing trajectories, correcting drifts and inaccuracies, and managing occlusions, where objects briefly disappear and reappear within the scene. By leveraging temporal coherence, the model maintains robust tracking despite these challenges. As shown in Fig. 4, the absence of trajectory refinement leads to decreased model performance due to blurring effects.

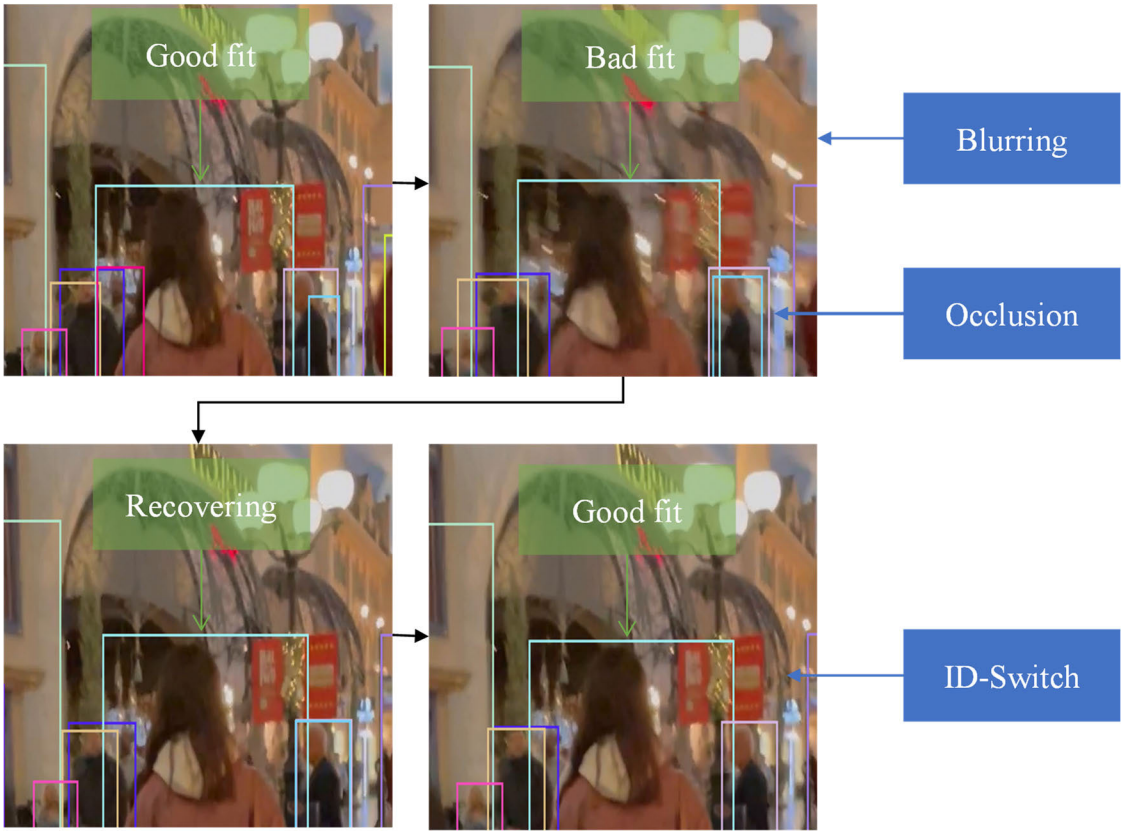


FIGURE 4. Severe blurring of the frames will result in a 'bad fit'. This is one type of error that is reduced by the proposed ATR module, which in return improves the overall performance.

The CFTformer model effectively handles occlusions by leveraging contextual information captured by the encoder. At each step, this information is passed to the ATR module along with the past tracking queries, where instant trajectories (pairwise trajectories) are computed for the t time components. These trajectories are then processed by a separate attention module, which refines them based on ground truth data. Simultaneously, the feature maps of the current frame, embedded with the three-dimensional positional encoding, is also passed to the ATR module, where the multi-head attention module utilizes these inputs to adjust the trajectories. From the refined trajectory, queries can be adjusted before being passed to the next decoder for tracking. This not only enables CFTformer to refine query trajectories, but also allows it to track the motion of past queries that may be temporarily occluded by other objects. Fig. 5 illustrates the structure of the proposed ATR module.

When an object reappears after occlusion, the model accurately re-identifies it and resumes tracking from its last known position. This minimizes identity switches and preserves trajectory integrity over time. As new frames are processed, the model iteratively updates trajectories, ensuring accurate tracking throughout the video sequence. The proposed ATR is a memory-based module that increases training time by approximately 20%; however, it has been shown to reduce identity switches and enhance overall performance.

G. OUTPUT GENERATION

Finally, the model generates the output, which includes trajectories of the tracked objects, updated bounding boxes, and object identities for each frame. Once the outputs are generated, the performance of the proposed CFTformer model is evaluated using several standard object tracking metrics. To compare the model's performance with other SOTA models, three key metrics are selected: identification F1 score (IDF1), multi-object tracking accuracy (MOTA), and multi-object tracking precision (MOTP).

1) IDF1 SCORE

IDF1 is one of the most important evaluation metrics for assessing the accuracy of MOT algorithms. It accounts for both the precision and recall of the detected trajectories, emphasizing the correctness of object identities across frames. This metric is particularly important as it penalizes identity switches, where an object's identity is incorrectly reassigned during tracking, and fragmented tracks, where a single object is represented by multiple tracks. Based on the F1 score, IDF1 is computed as the true positive matches between prediction and ground-truth, where each match must maintain the correct object identity [34], [35].

$$IDF1 = \frac{2 \times IDP \times IDR}{IDP + IDR} \tag{10}$$

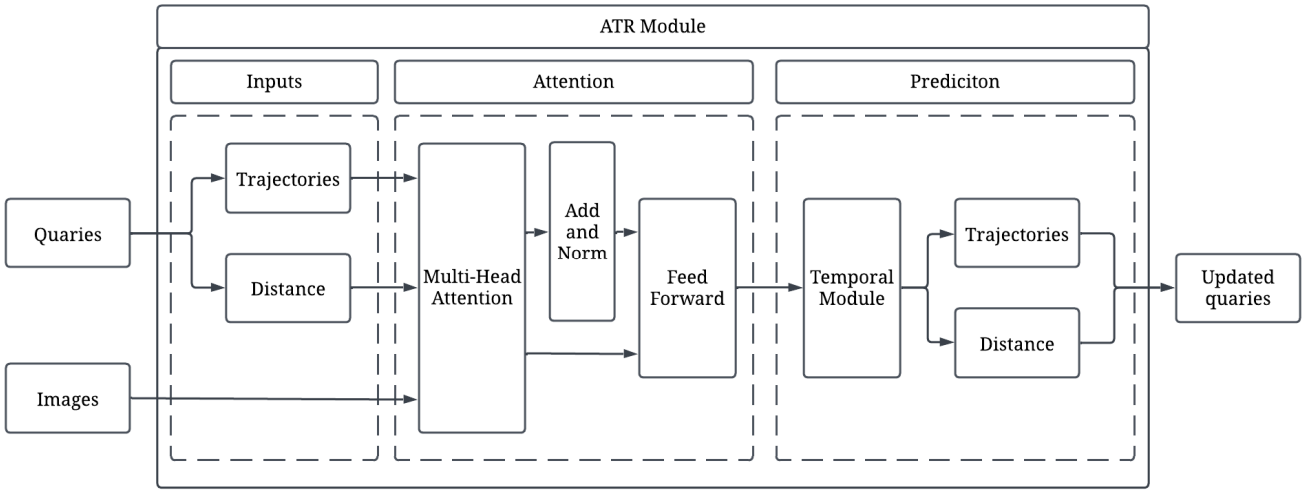


FIGURE 5. The proposed ATR module for improving performance drops caused by motion blurring.

In (10), ID precision (IDP) is calculated as the ratio of correctly identified detections to the total number of detected objects, while ID recall (IDR) is the ratio of correctly identified detections to the total number of ground-truth objects.

2) MOTA

MOTA is another widely used metric for evaluating the performance of MOT systems. It consolidates several error sources including false positives, false negatives, and identity switches, into a single measure. As a comprehensive metric, it provides a clear indication of how well a tracking system maintains object identities and accurately detects objects over time [43].

$$MOTA = 1 - \frac{FP + FN + IDS}{GT} \tag{11}$$

In (11), FP denotes the number of false positives, FN represents the number of false negatives, IDS indicates the number of identity switches, and GT refers to the total number of ground-truth objects.

3) MOTP

MOTP is a metric used to evaluate the spatial accuracy of tracking algorithms in MOT tasks. It measures how accurately the predicted bounding boxes align with their respective ground-truth positions over a sequence of frames. While MOTP is less commonly used than IDF1 and MOTA, it provides valuable insights into tracking performance. Using this metric alongside others allows for better understanding of the model's performance. MOTP is calculated by averaging the intersection over union (IoU) between each predicted bounding box and its corresponding ground-truth for all matched object pairs [43]. Essentially, MOTP quantifies the alignment between the predicted and actual locations offering insight into the positional accuracy of the tracking system.

$$MOTP = \frac{\sum_{i,t} d_{i,t}}{\sum_t C_t} \tag{12}$$

Equation (12) defines the calculation of the MOTP metric, where c_t is the number of matches between the predicted and ground-truth objects at time t and $d_{i,t}$ is the distance between ground truth object i and its corresponding predicted object at time t. This distance can be measured using the IoU of bounding boxes or the Euclidean distance between object centers.

H. DATASETS

To evaluate the proposed model, five datasets were employed. The first two, COCO and Crowdhuman, were used for pre-training, while MOT17, MOT20, and BDD100K were used for fine-tuning and performance evaluation.

1) COCO

The COCO dataset is a large-scale benchmark dataset widely used in computer vision for tasks such as object detection and tracking. Introduced by Lin et al. [42] in 2014, it holds over 200,000 labeled images with more than 1.5 million object instances spanning 80 object categories. A key characteristic of COCO is its emphasis on contextual information, featuring images that depict objects in natural environments with complex backgrounds and interactions.

2) CrowdHuman

The CrowdHuman dataset is a comprehensive benchmark designed for human detection and crowd analysis in dense and crowded scenarios. Introduced by Shao et al. [44] in 2018, it contains approximately 15,000 images with over 339,000 annotated human instances, making it one of the largest datasets dedicated to human detection.

TABLE 1. Comparison of performance of multi-object tracking of the proposed CFTformer model with the current SOTA models using the MOT17 test data with private detection protocol. Several standard metrics have been used to compare the top CNN-based and transformer-based approaches.

Methods	Year	Publication	IDF1	MOTA	MOTP	ID-S
CNN-based approaches						
CenterTrack [35]	2020	ECCV	64.7	67.8	-	3039
FairMOT [10]	2021	IJCV	72.3	73.7	-	3303
ByteTrack [36]	2022	ECCV	77.3	80.3	-	2196
BoT-SORT [37]	2022	arXiv	80.2	80.5	-	1212
OC-SORT [38]	2023	CVPR	77.5	78.0	-	1950
Transformer-based approaches						
TransTrack [23]	2020	arXiv	63.9	74.5	80.6	3663
TransCenter [39]	2022	TPAMI	62.2	73.2	81.1	4614
TrackFormer [20]	2022	CVPR	63.9	65.0	-	2829
MOTR [24]	2022	ECCV	68.6	73.4	-	2439
TransMOT [27]	2023	WACV	62.2	73.2	82.0	2346
MOTRv2 [25]	2023	CVPR	75.0	78.6	-	-
STDFormer [40]	2023	TCSVT	73.1	78.8	-	4998
ABQ-Track [42]	2023	Sensors	65.4	75.9	-	3135
ContrasTR [26]	2024	WACV	71.8	73.7	-	2619
DiffusionTrack [41]	2024	AAAI	73.8	77.9	-	3819
ETTrack [29]	2025	Applied Intelligence	75.9	79.0	-	2118
CFTformer (Proposed method)			77.6	76.5	79.4	2123

3) MOT17

The MOT17 dataset is a widely recognized benchmark in MOT, designed to evaluate tracking algorithms in challenging real-world scenarios. Introduced by Milan et al. [31] in 2016, it comprises a diverse set of video sequences recorded in various environments, including crowded public spaces and urban scenes. This dataset contains over 500,000 manually annotated bounding boxes across 14 different video sequences, each featuring multiple objects with unique identities.

4) MOT20

MOT20 dataset is a prominent benchmark designed for evaluating MOT algorithms, particularly in highly crowded and complex scenes. Released as part of the MOTChallenge, the MOT20 dataset includes eight video sequences recorded in challenging environments such as busy public spaces and dense pedestrian zones [44]. These sequences feature a total of over a million annotated bounding boxes, with thousands of unique identities.

5) BDD100K

The BDD100k dataset is a comprehensive large-scale dataset used for autonomous driving research, offering diverse annotations and data modalities. It includes 100,000 video clips captured in various weather conditions, times of day, and geographic locations, providing a rich source of information for training and testing autonomous driving models. The dataset includes annotations for tasks such as object detection, lane detection, and instance segmentation, making it a versatile tool for multi-task learning [45]. Additionally, BDD100k is one of the most diverse driving datasets, emphasizing the importance of generalization in real-world driving scenarios.

IV. RESULTS AND DISCUSSION

The proposed CFTformer is initialized with pre-trained weights, from deformable DETR model trained on the COCO dataset, leveraging transfer learning [34]. Before training, all datasets are preprocessed to ensure compatibility with the model. This process involves converting annotations to a standardized format, resizing frames to a consistent dimension, and normalizing pixel values. Annotations typically include bounding boxes and object identities for each frame, both essential for training and evaluation. The model is then retrained for 100 epochs on the CrowdHuman dataset, followed by an additional 250 epochs on the MOT17 dataset.

For all training procedures, a batch size of 1 was selected with a learning rate of 10^{-4} for the encoder/decoder and

TABLE 2. Comparison of the performance of the model with and without the use of ATR and CrowdHuman pre-training on MOT17 test data.

Methods	IDF1	MOTA	MOTP	ID-S
CFTformer	77.6	76.5	79.4	2123
w/o Trajectory refinement (ATR)	73.2	71.9	72.4	2734
w/o CrowdHuman pre-training	75.7	75.3	77.4	2582

TABLE 3. Comparison of performance of multi-object tracking of the proposed CFTformer model with the current SOTA models using the MOT20 test data with private detection protocol.

Methods	Year	Publication	IDF1	MOTA	MOTP	ID-S
CNN-based approaches						
FairMOT [10]	2021	IJCV	67.3	61.8	-	5243
ByteTrack [36]	2022	ECCV	75.2	77.8	-	1223
BoT-SORT [37]	2022	arXiv	77.5	77.8	-	1257
OC-SORT [38]	2023	CVPR	76.3	75.7	-	942
Transformer-based approaches						
TransTrack [23]	2020	arXiv	59.4	65.0	-	3608
TransCenter [39]	2022	TPAMI	50.4	61.9	-	4653
MOTRv2 [25]	2023	CVPR	72.2	76.2	-	-
STDFormer [40]	2023	TCSVT	72.1	76.2	-	3166
ABQ-Track [42]	2023	Sensors	60.3	66.3	-	3383
DiffusionTrack [41]	2024	AAAI	66.3	72.8	-	4117
CFTformer (Proposed method)			72.8	75.9	76.8	2745

TABLE 4. Comparison of the performance of the model with and without the use of ATR and CrowdHuman pre-training for MOT20 test data.

Methods	IDF1	MOTA	MOTP	ID-S
CFTformer	72.8	75.9	76.8	2745
w/o Trajectory refinement	69.9	71.2	73.7	3226
w/o CrowdHuman pre-training	70.6	73.6	75.1	3379

10^{-5} for the training of the backbone. All hyperparameters were carefully chosen to optimize performance. The model was trained using the WAdam optimizer on high-performance computing machine with two Nvidia RTX 3090 Ti GPUs to handle the high computational demand. The training process minimizes a composite loss function that includes components for detection accuracy and identity preservation. The loss functions ensures that the model learns to accurately detect objects while maintaining consistent identities over time. To enhance generalization capabilities, several data augmentation techniques, including random cropping and random flipping, are applied.

Table 1 compares the IDF1, MOTA, and MOTP metrics of the CFTformer model against top models from both CNN-based and transformer-based approaches. The results indicate that the proposed approach performs well compared to other models, achieving SOTA performance. CFTformer improves the IDF1 metric of the best-performing transformer-based model by 1.7%, while its MOTA score remains comparable to other top-performing models. Additionally, number of identity switches with CFTformer was second lowest among all transformer-based models in Table 1. The MOTP metric was not reported in many cases. Some newer CNN-based models continue to achieve high performance; however, with further research into transformer-based approaches and novel techniques to enhance robustness, transformer models are expected to see significant improvements in the future. Table 2 presents a comparison of the effects of trajectory refinement and pre-training on the CrowdHuman dataset. While both significantly impact tracking performance, trajectory refinement appears to have a greater effect.

Similarly, CFTformer was trained on the MOT20 dataset for further comparison. Table 3 presents a comparison of

TABLE 5. Comparison of performance of multi-object tracking of the proposed CFTformer model with the current SOTA models using the BDD100K validation data.

Methods	Year	Publication	IDF1	MOTA	MOTP	ID-S
CNN-based approaches						
Yu *et al.* [45]	2018	arXiv	-	56.7	84.1	9098
QDTrack [46]	2023	TPAMI	68.2	73.3	-	8478
Unicorn [47]	2022	ECCV	71.3	66.6	-	10876
Transformer-based approaches						
MOTR [24]	2022	ECCV	65.8	56.2	-	-
MOTRv2 [25]	2023	CVPR	65.6	72.7	-	-
ContrasTR [26]	2024	WACV	56.5	42.8	-	10793
CFTformer (Proposed method)			67.7	59.8	69.7	5273

TABLE 6. Comparison of the performance of the model with and without the use of ATR for BDD100K validation data.

Methods	IDF1	MOTA	MOTP	ID-S
CFTformer	67.7	59.8	69.7	5273
w/o Trajectory refinement	65.4	52.3	65.3	8752

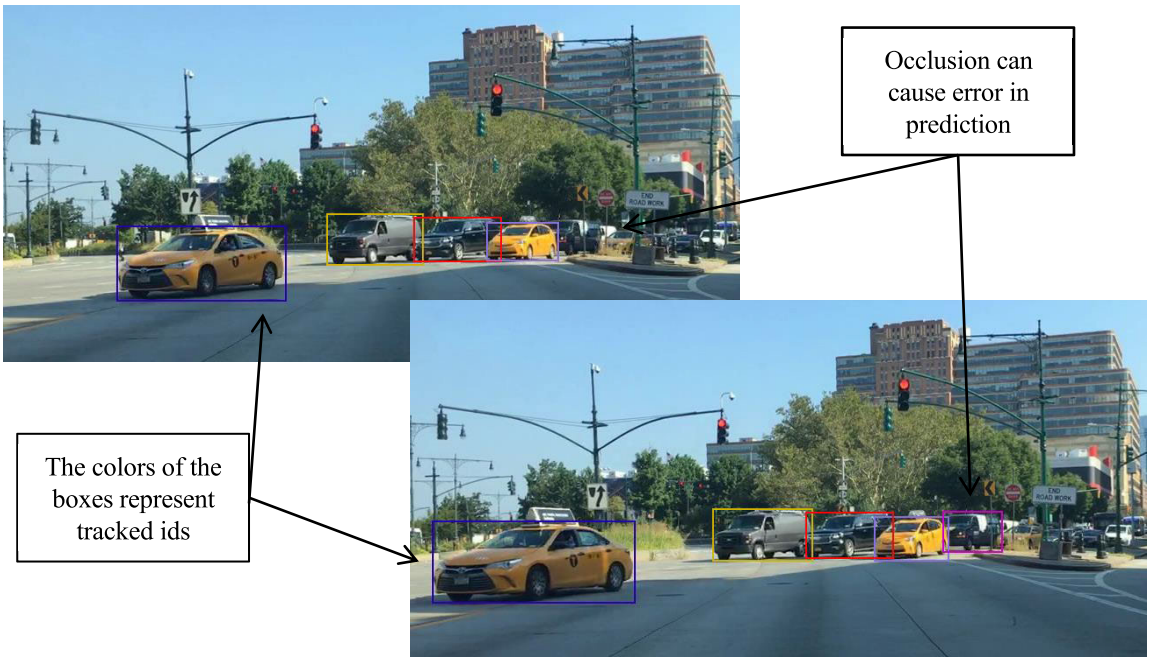


FIGURE 6. An example of tracking across frames using the CFTformer method for the BDD100K dataset.

similar models on MOT20. The high density and frequent occlusions in MOT20 make maintaining accurate and consistent tracking more challenging, which is why overall performance metrics tend to be lower compared to MOT17. However, CFTformer still achieves SOTA performance on MOT20 as well. This model recorded the highest IDF1 score among transformer-based trackers while having the lowest number of identity switches on the MOT20 test set. Table 4 highlights the primary reason for this improvement. As shown, the ATR module plays a crucial role in reducing identity switches compared to similar approaches. By utilizing the three-dimensional positions of the current

and previous frames to correct the calculated trajectories of tracked objects, this module significantly reduces identity switches, thereby improving tracking performance.

Aside from the MOT17 and MOT20 datasets, CFTformer was also trained and evaluated on the BDD100K dataset. Table 5 presents a comparison of the MOT results achieved by the proposed model against current SOTA approaches on this large-scale dataset. While the proposed model has a lower MOTA, it achieves a higher IDF1 score compared to other transformer-based approaches.

The CNN-based methods introduced by Yu et al. [45] and Fisher et al. [46] achieve higher MOTA, however transformer-based models tend to have fewer identity switches. Table 6 demonstrates the effect of the ATR module, showing its contribution to performance improvement, particularly in reducing identity switches. This makes it a valuable component for tracking applications where identity preservation is crucial. Fig. 6 illustrates an example of object tracking across two consecutive frames of the BDD100K dataset. It highlights how even partial occlusion can prevent the algorithm from detecting objects. When objects remain consistently obscured, tracking errors accumulate, leading to more identity switches. The proposed ATR module helps mitigate such errors, effectively reducing identity switches in challenging scenarios.

V. CONCLUSION AND FUTURE WORK

The proposed CFTformer is an end-to-end MOT method that leverages three-dimensional tracking of queries across frames. To achieve this, a novel positional encoding scheme and a redesigned encoder-decoder structure were required. This approach has demonstrated SOTA performance, matching or even surpassing previous methods in certain metrics.

Additionally, a trajectory refinement module based on the attention mechanism was developed to track the movement of bounding boxes and mitigate errors caused by motion blur in some frames. The proposed ATR module and pre-training strategy were found to have a significant impact on model performance. The key assumption behind trajectory refinement is that objects move at a slower pace relative to the frame rate, allowing the proposed algorithm to refine the relative position of the camera with respect to the object. This module is particularly useful in AVs, where accurately tracking an object's relative trajectory and speed is critical within a defined range.

The use of deformable DETR and bipartite matching played a crucial role in lowering the computational load of the model. However, this algorithm still requires large datasets and significant computational resources for training. While employing a pre-trained backbone model alleviates some of these challenges, the high computational cost remains a key limitation of transformer-based tracking approaches. Self-supervised training on large-scale uncurated data presents a potential solution to address this issue. The increase in computational capacity of new hardware and the development of more efficient architectures are expected to drive further improvements in this field.

REFERENCES

[1] Y. Xu, X. Zhou, S. Chen, and F. Li, "Deep learning for multiple object tracking: A survey," *IET Comput. Vis.*, vol. 13, no. 4, pp. 355–368, Jun. 2019, doi: 10.1049/iet-cvi.2018.5598.

[2] R. Yao, G. Lin, S. Xia, J. Zhao, and Y. Zhou, "Video object segmentation and tracking: A survey," *ACM Trans. Intell. Syst. Technol. (TIST)*, vol. 11, no. 4, pp. 1–47, Jan. 2019.

[3] Z. Sun, J. Chen, L. Chao, W. Ruan, and M. Mukherjee, "A survey of multiple pedestrian tracking based on tracking-by-detection framework," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 31, no. 5, pp. 1819–1833, May 2021.

[4] A. Yilmaz, O. Javed, and M. Shah, "Object tracking: A survey," *ACM Comput. Surv.*, vol. 38, no. 4, p. 13, 2006.

[5] S. M. Marvasti-Zadeh, L. Cheng, H. Ghanei-Yakhdan, and S. Kasaei, "Deep learning for visual tracking: A comprehensive survey," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 5, pp. 3943–3968, May 2022.

[6] L. Bertinetto, J. Valmadre, J. F. Henriques, A. Vedaldi, and P. H. Torr, "Fully-convolutional Siamese networks for object tracking," in *Proc. ECCV*, Amsterdam, The Netherlands. Cham, Switzerland: Springer, Oct. 2016, pp. 850–865.

[7] D. Held, S. Thrun, and S. Savarese, "Learning to track at 100 FPS with deep regression networks," in *Proc. ECCV*, Amsterdam, The Netherlands. Cham, Switzerland: Springer, Oct. 2016, pp. 749–765.

[8] W. Luo, J. Xing, A. Milan, X. Zhang, W. Liu, and T.-K. Kim, "Multiple object tracking: A literature review," *Artif. Intell.*, vol. 293, Apr. 2021, Art. no. 103448, doi: 10.1016/j.artint.2020.103448.

[9] N. Wojke, A. Bewley, and D. Paulus, "Simple online and realtime tracking with a deep association metric," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2017, pp. 3645–3649.

[10] Y. Zhang, C. Wang, X. Wang, W. Zeng, and W. Liu, "FairMOT: On the fairness of detection and re-identification in multiple object tracking," *Int. J. Comput. Vis.*, vol. 129, no. 11, pp. 3069–3087, Nov. 2021.

[11] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, Jun. 2017, pp. 5998–6008.

[12] A. B. Amjoud and M. Amrouch, "Object detection using deep learning, CNNs and vision transformers: A review," *IEEE Access*, vol. 11, pp. 35479–35516, 2023.

[13] K. Han, A. Xiao, E. Wu, J. Guo, C. Xu, and Y. Wang, "Transformer in transformer," in *Proc. Adv. Neural Inf. Process. Syst.*, Jan. 2021, pp. 15908–15919.

[14] J. Devlin, M. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. NAACL-HLT*, Jan. 2018, p. 2.

[15] A. Radford, J. Wu, R. Child, D. Luan, D. Amodei, and I. Sutskever, "Language models are unsupervised multitask learners," *OpenAI Blog*, vol. 1, no. 8, p. 9, 2019.

[16] Y. Wang, A. Mohamed, D. Le, C. Liu, A. Xiao, J. Mahadeokar, H. Huang, A. Tjandra, X. Zhang, F. Zhang, C. Fuegen, G. Zweig, and M. L. Seltzer, "Transformer-based acoustic modeling for hybrid speech recognition," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2020, pp. 6874–6878.

[17] G. M. Correia, V. Niculae, and A. F. T. Martins, "Adaptively sparse transformers," in *Proc. Conf. Empirical Methods Natural Lang. Process. 9th Int. Joint Conf. Natural Lang. Process. (EMNLP-IJCNLP)*, 2019, pp. 2174–2184.

[18] S. W. Zamir, A. Arora, S. Khan, M. Hayat, F. S. Khan, and M. Yang, "Restormer: Efficient transformer for high-resolution image restoration," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 5718–5729.

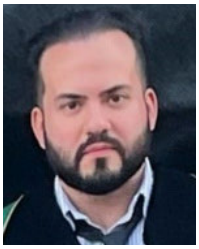
[19] Z. Wang, H. Zhao, Y. Li, S. Wang, P. H. S. Torr, and L. Bertinetto, "Do different tracking tasks require different appearance models," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, Dec. 2021, pp. 726–738.

[20] T. Meinhardt, A. Kirillov, L. Leal-Taixé, and C. Feichtenhofer, "TrackFormer: Multi-object tracking with transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 8834–8844.

[21] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.
[22] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, Jan. 2020, pp. 213–229.
[23] P. Sun, J. Cao, Y. Jiang, R. Zhang, E. Xie, Z. Yuan, C. Wang, and P. Luo, "TransTrack: Multiple object tracking with transformer," 2020, *arXiv:2012.15460*.
[24] F. Zeng, B. Dong, Y. Zhang, T. Wang, X. Zhang, and Y. Wei, "MOTR: End-to-end multiple-object tracking with transformer," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, Jan. 2022, pp. 659–675.
[25] Y. Zhang, T. Wang, and X. Zhang, "MOTRv2: Bootstrapping end-to-end multi-object tracking by pretrained object detectors," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 22056–22065.
[26] P.-F.-D. Plaen, N. Marinello, M. Proesmans, T. Tuytelaars, and L. Van Gool, "Contrastive learning for multi-object tracking with transformers," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2024, pp. 6853–6863.
[27] P. Chu, J. Wang, Q. You, H. Ling, and Z. Liu, "TransMOT: Spatial–temporal graph transformer for multiple object tracking," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2023, pp. 4859–4869.
[28] S. Chen, P. Sun, Y. Song, and P. Luo, "DiffusionDet: Diffusion model for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 19773–19786.
[29] X. Han, N. Oishi, Y. Tian, E. Ucurum, R. Young, C. Chatwin, and P. Birch, "ETTrack: Enhanced temporal motion predictor for multi-object tracking," *Appl. Intell.*, vol. 55, no. 1, pp. 1–17, Jan. 2025.
[30] A. Bewley, Z. Ge, L. Ott, F. Ramos, and B. Upcroft, "Simple online and realtime tracking," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2016, pp. 3464–3468.
[31] A. Milan, L. Leal-Taixe, I. Reid, S. Roth, and K. Schindler, "MOT16: A benchmark for multi-object tracking," 2016, *arXiv:1603.00831*.
[32] P. Jiang, D. Ergu, F. Liu, Y. Cai, and B. Ma, "A review of YOLO algorithm developments," *Proc. Comput. Sci.*, vol. 199, pp. 1066–1073, Jan. 2022.
[33] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[34] X. Zhu, W. Su, L. Lu, B. Li, X. Wang, and J. Dai, "Deformable DETR: Deformable transformers for end-to-end object detection," in *Proc. Int. Conf. Learn. Represent.*
[35] X. Zhou, V. Koltun, and P. Krähenbühl, "Tracking objects as points," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, Jan. 2020, pp. 474–490.
[36] Y. Zhang, P. Sun, Y. Jiang, D. Yu, F. Weng, Z. Yuan, P. Luo, W. Liu, and X. Wang, "ByteTrack: Multi-object tracking by associating every detection box," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, Jan. 2022, pp. 1–21.
[37] N. Aharon, R. Orfaig, and B.-Z. Bobrovsky, "BoT-SORT: Robust associations multi-pedestrian tracking," 2022, *arXiv:2206.14651*.
[38] J. Cao, J. Pang, X. Weng, R. Khirodkar, and K. Kitani, "Observation-centric SORT: Rethinking SORT for robust multi-object tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 9686–9696.
[39] Y. Xu, Y. Ban, G. Delorme, C. Gan, D. Rus, and X. Alameda-Pineda, "TransCenter: Transformers with dense representations for multiple-object tracking," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 6, pp. 7820–7835, Jun. 2023.
[40] M. Hu, X. Zhu, H. Wang, S. Cao, C. Liu, and Q. Song, "STDFormer: Spatial–temporal motion transformer for multiple object tracking," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 11, pp. 6571–6594, Nov. 2023.
[41] R. Luo, Z. Song, L. Ma, J. Wei, W. Yang, and M. Yang, "DiffusionTrack: Diffusion model for multi-object tracking," in *Proc. AAAI Conf. Artif. Intell.*, Mar. 2024, vol. 38, no. 5, pp. 3991–3999.
[42] Q. Wang, C. Lu, L. Gao, and G. He, "Transformer-based multiple-object tracking via anchor-based-query and template matching," *Sensors*, vol. 24, no. 1, p. 229, Dec. 2023.
[43] K. Bernardin and R. Stiefelhagen, "Evaluating multiple object tracking performance: The CLEAR MOT metrics," *EURASIP J. Image Video Process.*, vol. 2008, pp. 1–10, Jan. 2008.
[44] S. Shao, Z. Zhao, B. Li, T. Xiao, G. Yu, X. Zhang, and J. Sun, "CrowdHuman: A benchmark for detecting human in a crowd," 2018, *arXiv:1805.00123*.
[45] F. Yu, W. Xian, Y. Chen, F. Liu, M. Liao, V. Madhavan, and T. Darrell, "Bdd100k: A diverse driving video database with scalable annotation tooling," 2018, *arXiv:1805.04687*.
[46] T. Fischer, T. E. Huang, J. Pang, L. Qiu, H. Chen, T. Darrell, and F. Yu, "QDTrack: Quasi-dense similarity learning for appearance-only multiple object tracking," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 12, pp. 15380–15393, Dec. 2023.
[47] B. Yan, Y. Jiang, P. Sun, D. Wang, Z. Yuan, P. Luo, and H. Lu, "Towards grand unification of object tracking," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, Jan. 2022, pp. 733–751.



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